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Hakimi Heris

CONTACT  
INFORMATION

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RESEARCH  
INTERESTS

My research builds the layer between quantum algorithms and the devices that run them: hardware-native instruction sets and the universality results that justify them, the compilers and control pulses that target them, and the trapped-ion and photonic hardware they execute on. I work on quantum algorithms and QSP/QSVT, quantum error analysis and fault tolerance, hardware–algorithm co-design, adaptive quantum control, hybrid CV–DV computation, photonic quantum computing, cavity QED, and spin–photon interfaces. My aim is to make early fault-tolerant quantum computation practical by co-designing the algorithm, the architecture, and the control together rather than in isolation.

My work falls into four threads, with a representative result in each:

- **Instruction set architectures for hybrid quantum computation** [P2]  
A native trapped-ion qudit–oscillator ISA whose three core primitives generate universal Hamiltonians, with no synthesized mode-coupling operation required; validated on dissipative vibronic dynamics compiled onto a single ququart.
- **Pulse-level control and compilation for trapped ions** [C4][U2][U3][P1]  
*HyPulse*: a GRAPE-based pulse-synthesis framework for hybrid CV–DV qubit–oscillator gates, with a reusable library of compound pulse gadgets for Hamiltonian simulation.
- **Quantum signal processing: robustness and photonic realization** [J1][P3][P4]  
Showed that block-encoding and phase-angle errors cross-compensate to first order, and built a constrained optimizer suppressing correlated control error by 4–19×; co-led the first experimental realization of QSP on a programmable photonic circuit.
- **Nanophotonics and quantum materials** [C2][C3][J2][J3]  
Inverse-designed nanophotonic cavities for spin–photon interfacing, and MEMS/NEMS platforms for optical strain studies of 2D materials.

EDUCATION

**Ph.D. Candidate, Electrical Engineering**, North Carolina State University 2024–Present  
Advisor: Prof. Yuan Liu · Expected May 2027 · GPA: 4.0/4.0  
Dissertation: *Co-Design of Quantum Algorithms, Architectures, and Control for Early Fault-Tolerant Hardware*

**M.S., Electrical Engineering**, Montana State University 2022–2024  
Advisor: Prof. David L. Dickensheets · GPA: 3.87/4.0  
Thesis: *MEMS Cantilevers for Dynamic Strain Studies of 2D Materials*

**M.S., Electrical Engineering (Electronics)**, Qazvin Azad University, Iran 2012–2015

**B.S., Electrical Engineering (Electronics)**, Karaj Azad University, Iran 2006–2011

PUBLICATION  
SUMMARY

**Published:** 7 peer-reviewed papers, 2 of them first-authored. **In progress:** 3 preprints or manuscripts under review and 4 in preparation, 5 of these 7 first- or co-first-authored.  
**Impact:** 90 citations, h-index 3, i10-index 2 (Google Scholar, August 2026).  
Venues include IEEE QCE, HPCA (under review), ISCA workshops, SPIE Photonics for Quantum, and *APL Computational Physics*.

*Journal Articles*

- [J3] E. Karooby, H. Sahbafar, **M. Hakimi Heris**, A. Hadi, V. Eskandari. “Identification of Low Concentrations of Flucytosine Drug Using a SERS-Active Filter Paper Substrate.” *Plasmonics* 19(2), 855–863, 2024.
- [J2] V. Eskandari, H. Sahbafar, E. Karooby, **M. Hakimi Heris**, S. Mehmandoust, D. Razmjoue, A. Hadi. “SERS Filter Paper Substrates Decorated with Silver Nanoparticles for the Detection of Molecular Vibrations of Acyclovir Drug.” *Spectrochimica Acta Part A* 298, 122762, 2023.
- [J1] K. J. Joven, E. R. Das, J. Bierman, A. Majumdar, **M. Hakimi Heris**, Y. Liu. “Scalable Quantum Computational Science: A Perspective from Block-Encodings and Polynomial Transformations.” *APL Computational Physics* 2, 010901, 2026.

*Conference Papers*

- [C4] **M. Hakimi Heris**, Y. Liu, F. Mueller. “HyPulse: A Pulse Synthesis Framework for Hybrid Qubit–Oscillator Gates on Trapped-Ion Platform.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026. arXiv:2604.26804.
- [C3] E. Karooby, **M. Hakimi Heris**, Q. Gu, D. Farfurnik. “Inverse Design of Non-Periodic Circular Cavities for Efficient Spin–Photon Interfacing by Genetic Algorithm.” *SPIE Optical Modeling and Performance Predictions XV* 13594, 116–119, 2025.
- [C2] A. Das, N. K. Vij, **M. Hakimi Heris**, D. Farfurnik. “Optimization of Circular Gratings via Guided-Mode-Expansion-Method-Based Inverse Design.” *SPIE Photonics for Quantum* 13563, 65–71, 2025.
- [C1] **M. Hakimi Heris**, R. Faez. “Investigation of the Possibility of Using Single-Electron Transistors in Digital-to-Time Converters.” *IEEE Global Congress on Electrical Engineering (GCElecEng)*, 150–153, 2021.

UNDER REVIEW  
AND PREPRINTS

- [U3] C. Chu, **M. Hakimi Heris**, H. Zhou, Y. Liu. “FACT: A Fidelity-Aware Compiler for Trapped-Ion Quantum Computers.” Under review, *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, 2027.
- [U2] R. Patel, **M. Hakimi Heris**, Y. Liu, F. Mueller. “Synthesizing Compound Pulse Gadgets for Hamiltonian Simulation on Trapped-Ion Platforms.” arXiv:2607.00826, 2026.
- [U1] **M. Hakimi Heris**. “Graph State Generation Based on Quantum Photonic Devices, Enabling Measurement-Based Quantum Computing.” arXiv:2607.16667, 2026.

MANUSCRIPTS IN  
PREPARATION

- [P4] E. Karooby<sup>\*</sup>, **M. Hakimi Heris**<sup>\*</sup>, Y. Liu, Q. Gu. “Universal Function Approximation Using Quantum Signal Processing on a Photonic Integrated Platform.” <sup>\*</sup> Equal contribution.
- [P3] **M. Hakimi Heris**, [co-authors], Y. Liu. “Robustness-Aware Quantum Signal Processing: Error Cancellation, Optimization, and Learned Angle Synthesis.”
- [P2] **M. Hakimi Heris**, Y. Liu, F. Mueller. “A Native Trapped-Ion Hybrid Qudit–Oscillator Instruction Set.”
- [P1] **M. Hakimi Heris**, Y. Liu, F. Mueller. “Learned Pulse Synthesis for Hybrid CV–DV Gates on Trapped-Ion Hardware.”

FUNDING AND PROJECTS

Grand Unification of Quantum Algorithms on Photonic Integrated Circuits

- Source of Support: NSF I-Corps Teams (Quantum+X)
- Role: Co-Entrepreneurial Lead
- Project Dates: 2026
- PI: Qing Gu (NC State, ECE/Physics); Technical Co-Lead: Yuan Liu (NC State, ECE/CS); Co-Entrepreneurial Lead: Elaheh Karooby; Industry Mentor: Amirhassan Shams-Ansari (Monarch Quantum)
- Total Project Amount: \$50,000

NQVL:QTSD:Pilot/Design: Quantum Advantage-Class Trapped Ion System (QACTI)

- Source of Support: NSF National Quantum Virtual Laboratory, via subcontract from Duke University
- Role: Supported as Graduate Research Assistant
- Lead PI: Ken Brown (Duke University); NC State: Yuan Liu, Frank Mueller
- Phase 1 (NSF 2410675): \$100,000, 08/2024–07/2025
- Phase 2 (NSF 2531350): \$390,000, 09/2025–08/2027
- Building an advantage-class trapped-ion quantum computer for remote use by the scientific community, with an open-source hardware/software stack.

*[Also acknowledged in your papers: NSF PHY-2325080, OMA-2120757, and DOE DE-SC0025384. Add if they support your position.]*

SELECTED PRESENTATIONS

- “Experimental Realization of Quantum Signal Processing on a Photonic Integrated Circuit,” IEEE Photonics Conference. 2026
- “Toward Compact Multiqubit Gates on Trapped Ions: A Hardware-Aware Pulse Synthesis,” QCCC-26 Workshop, ISCA. 2026
- “Simulations of Nonadiabatic Vibronic Dynamics on Hybrid Qudit–Qumode Trapped-Ion Devices,” RQS Institute Workshop, University of Maryland. 2026
- “MEMS Devices for In-Situ Straining of 2D Materials,” MonArk NSF Quantum Foundry Site Visit, Bozeman, MT. 2024
- “MEMS Cantilevers for Dynamic Strain Studies of 2D Materials,” Optical Science & Engineering Conference (OpTeC), Bozeman, MT. 2023

AWARDS AND PROFESSIONAL PROGRAMS

- Prospects in Theoretical Physics (PiTP), Institute for Advanced Study, Princeton 2026
- Graduate Merit Award, College of Engineering, NC State University 2024–2025
- AIM Photonics Summer Academy, MIT 2023
- Summer of Quantum Short Course, LPS Qubit Collaboratory 2023
- Excellent Graduate Student in Electronics Engineering, Qazvin Azad University 2015
- Top 3 Undergraduate Student in Electronics Engineering, Karaj Azad University 2009
- Best Patent, Kharazmi Festival, Tehran, Iran 2004

RESEARCH  
EXPERIENCE

**Graduate Researcher**, Quantum Computing and Engineering, NC State University2024–Present  
Advisor: Prof. Yuan Liu

- Designed a native trapped-ion hybrid qudit–oscillator instruction set and proved that three of its primitives generate universal Hamiltonians, with no synthesized mode-coupling operation required.
- Compiled dissipative vibronic dynamics of a three-chromophore complex onto a single  $^{171}\text{Yb}^+$  ququart, cutting gate count from 96 to 28 per Trotter step relative to the superconducting realization and reproducing published trajectories to  $\sim 1.1\%$  RMSE.
- Built HyPulse, a GRAPE-based pulse-synthesis framework for hybrid CV–DV gates, and extended HybridLane (PennyLane) to native qudit–oscillator circuits.
- Co-led the first experimental realization of QSP on a programmable photonic integrated circuit, synthesizing nonlinear functions on linear-optical hardware.
- Optical characterization and design of nanophotonic structures for light–matter interaction (cavity QED).
- Collaborations across ECE, physics, and computer science, including Duke University’s NQVL for trapped-ion experimental validation.

**Graduate Researcher**, MonArk NSF Quantum Foundry, Montana State University 2022–2024  
Advisors: Prof. David L. Dickensheets, Prof. Nicholas J. Borys

- Designed, fabricated, and characterized MEMS/NEMS devices for optical studies of 2D quantum materials, from EBL and photolithography through SEM/AFM metrology.

TEACHING  
EXPERIENCE

**Preparing the Professoriate (PTP) Fellow**, NC State University 2026–2027  
Year-long teaching apprenticeship: co-teach a course with a faculty mentor, develop a teaching portfolio and philosophy statement, and complete pedagogy coursework.

Teaching Assistant, Signals, Circuits, and Systems Laboratory, NC State University [term]  
Teaching Assistant, Microfabrication and Electrical Circuits Laboratories, Montana State University 2022–2023

*[Add enrollments and what you ran. For faculty applications, also list the courses you are prepared to teach.]*

MENTORING

**Samuel Firmansyah** — [level], NC State University Spring 2026  
Project: AI-based robust quantum signal processing (learned robust angle synthesis).

**Nidhi Grover** — [level], NC State University Spring 2026  
Project: AI-based pulse synthesis for hybrid CV–DV gates on trapped-ion hardware.

PROFESSIONAL  
SERVICE

Peer Review

Sub-reviewer on program-committee and editorial assignments for Prof. Yuan Liu and Prof. Frank Mueller (5 manuscripts, 2025–2026):

- *IEEE Journal of Selected Topics in Signal Processing*, Special Issue on Quantum Technologies — 1 manuscript (2026)
- IEEE International Conference on Quantum Computing and Engineering (QCE, IEEE Quantum Week) — 2 manuscripts, Quantum Applications (QAPP) and Quantum Machine Learning tracks (2026)
- International Workshop on Software Frameworks and Workload Management on Quantum and HPC Ecosystems (SFWM), at SC — 2 manuscripts (2025, 2026)

Memberships

[IEEE, APS, Optica, ACM — list any current student memberships.]

Departmental and Outreach

[Seminar or reading-group organizing, poster judging, recruiting/visit-day hosting, graduate student association, K–12 or undergraduate outreach. Anything counts; this subsection is still empty.]

TECHNICAL  
SKILLS

**Quantum algorithms & theory:** QSP/QSVT, block encodings, Hamiltonian simulation, open quantum systems, hybrid CV–DV computation, quantum error analysis.

**Quantum control:** GRAPE, pulse synthesis, constrained optimization, differentiable simulation, hardware-aware compilation, Hamiltonian-based control.

**Software:** Python, JAX, QuTiP, PennyLane, Qiskit, Strawberry Fields, MATLAB, Julia, C/C++.

**Machine learning:** Neural networks, Fourier neural operators, learned pulse synthesis, physics-informed and differentiable training.

**Photonics & hardware:** Programmable photonic circuits, Lumerical FDTD/INTERCONNECT/qINTERCONNECT/COMSOL, nanophotonic design and fabrication, optical characterization.

EARLIER INDUSTRY  
EXPERIENCE

Software and electrical engineering roles, 2011–2021 (FDM4 Canada; power and industrial systems engineering and project management).